A product minus a rectangle stays connected
Munkres, 23.9

Theorem 1. *Let X and Y be connected spaces, and let A be a proper subset of X and B a proper subset of Y. Then $(X \times Y) \setminus (A \times B)$ is connected.*

Proof. Let

$$S = (X \times Y) \setminus (A \times B).$$

Since A is a proper subset of X, there is a point $a_0 \in X$ with $a_0 \notin A$. Similarly, since B is a proper subset of Y, there is a point $b_0 \in Y$ with $b_0 \notin B$. Then $(a_0, b_0) \notin A \times B$, so $(a_0, b_0) \in S$; in particular, S is nonempty. It remains to show that S is preconnected, for a nonempty preconnected set is connected.

For $a \in X$, consider the vertical slice

$$V_a = \{a\} \times Y,$$

and for $b \in Y$, consider the horizontal slice

$$H_b = X \times \{b\}.$$

Each V_a is preconnected, since it is the product of the connected set $\{a\}$ with the connected set Y. Similarly, each H_b is preconnected, since it is the product of the connected set X with the connected set $\{b\}$.

Moreover, these slices lie in S when the fixed coordinate avoids the corresponding removed set. Indeed, if $a \notin A$, then for every $y \in Y$ the point (a, y) cannot lie in $A \times B$, since its first coordinate is not in A. Hence

$$V_a \subseteq S \qquad \text{whenever } a \notin A.$$

Likewise, if $b \notin B$, then for every $x \in X$ the point (x, b) cannot lie in $A \times B$, so

$$H_b \subseteq S \qquad \text{whenever } b \notin B.$$

Now let u and v be open subsets of $X \times Y$ such that

$$S \subseteq u \cup v,$$

and suppose that both $S \cap u$ and $S \cap v$ are nonempty. To prove that S is preconnected, we must show that $S \cap (u \cap v)$ is nonempty.

Suppose, toward a contradiction, that $S \cap (u \cap v)$ is empty. We use the following auxiliary observation. If $T \subseteq S$ is preconnected and T meets u, then $T \subseteq u$. Indeed, let $q \in T$. Since $T \subseteq S \subseteq u \cup v$, either $q \in u$ or $q \in v$. If $q \in v$, then T meets both u and v, while $T \subseteq u \cup v$; since T is preconnected, $T \cap (u \cap v)$ would be nonempty. A point in this intersection belongs to $T \subseteq S$ and to $u \cap v$, contradicting the assumption that $S \cap (u \cap v)$ is empty. Therefore $q \notin v$, so $q \in u$, and hence $T \subseteq u$. The same argument with the roles of u and v exchanged shows that if $T \subseteq S$ is preconnected and T meets v, then $T \subseteq v$.

We now show that $(a_0, b_0) \in u$. Choose a point $(x_u, y_u) \in S \cap u$. Since $(x_u, y_u) \notin A \times B$, we must have $x_u \notin A$ or $y_u \notin B$.

If $x_u \notin A$, then $V_{x_u} \subseteq S$ and V_{x_u} is preconnected. It meets u at (x_u, y_u), so by the auxiliary observation $V_{x_u} \subseteq u$. In particular, $(x_u, b_0) \in u$. Since $b_0 \notin B$, also $H_{b_0} \subseteq S$, and H_{b_0} is preconnected and meets u at (x_u, b_0). Hence $H_{b_0} \subseteq u$, and therefore $(a_0, b_0) \in u$.

If instead $y_u \notin B$, then $H_{y_u} \subseteq S$ and H_{y_u} is preconnected. It meets u at (x_u, y_u), so $H_{y_u} \subseteq u$. In particular, $(a_0, y_u) \in u$. Since $a_0 \notin A$, also $V_{a_0} \subseteq S$, and V_{a_0} is preconnected and meets u at (a_0, y_u). Hence $V_{a_0} \subseteq u$, and therefore $(a_0, b_0) \in u$.

Thus in either case $(a_0, b_0) \in u$.

Similarly, we show that $(a_0, b_0) \in v$. Choose a point $(x_v, y_v) \in S \cap v$. Since $(x_v, y_v) \notin A \times B$, either $x_v \notin A$ or $y_v \notin B$. In the first case, $V_{x_v} \subseteq S$ is preconnected and meets v, so $V_{x_v} \subseteq v$ by the auxiliary observation; hence $(x_v, b_0) \in v$, and then $H_{b_0} \subseteq S$ is preconnected and meets v, so $H_{b_0} \subseteq v$, giving $(a_0, b_0) \in v$. In the second case, $H_{y_v} \subseteq S$ is preconnected and meets v, so $H_{y_v} \subseteq v$; hence $(a_0, y_v) \in v$, and then $V_{a_0} \subseteq S$ is preconnected and meets v, so $V_{a_0} \subseteq v$, again giving $(a_0, b_0) \in v$.

Therefore $(a_0, b_0) \in S \cap (u \cap v)$, contradicting the assumption that this intersection is empty. It follows that $S \cap (u \cap v)$ is nonempty, so S is preconnected. Since S is also nonempty, S is connected. That is, $(X \times Y) \setminus (A \times B)$ is connected. $\square$